

고급통계분석 - Assignment02

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Problem 2.

(a).

```
gaussian_kernel_mat <- function(x, rho = 1){  
  x <- as.numeric(x)  
  outer(x, x, function(a, b) exp(-rho * (a - b)^2))  
}
```

(b).

```
library(Rcpp)  
  
cppFunction('  
#include <Rcpp.h>  
using namespace Rcpp;  
  
// [[Rcpp::export]]  
NumericMatrix gaussian_kernel_mat_cpp(const NumericVector& x, const  
double rho = 1.0){  
  int n = x.size();  
  NumericMatrix K(n, n);  
  for (int i = 0; i<n; ++i){  
    K(i, i) = 1.0;  
    for(int j = i+1; j<n; ++j){  
      double d = x[i] - x[j];  
      double d2 = d*d;  
      double kij = std::exp(-rho * d2);  
      K(i, j) = kij;  
      K(j, i) = kij;  
    }  
  }  
  return K;  
}  
' )
```

```
## Warning: No function found for Rcpp::export attribute at  
filee00c24b647d.cpp:5
```

(c).

```
set.seed(123)  
n <- 1000  
x <- runif(n)  
  
times_r <- replicate(50, system.time(gaussian_kernel_mat(x,  
1)))[['elapsed']])
```

```
times_rcpp <- replicate(50, system.time(gaussian_kernel_mat_cpp(x,  
1)))[['elapsed']])
```

```
mean(times_r); mean(times_rcpp)
```

```
## [1] 0.1198
```

```
## [1] 0.063
```

(d).

```
result <- data.frame(  
  Method = c('R', 'Rcpp'),  
  mean_time = c(mean(times_r), mean(times_rcpp))  
)
```

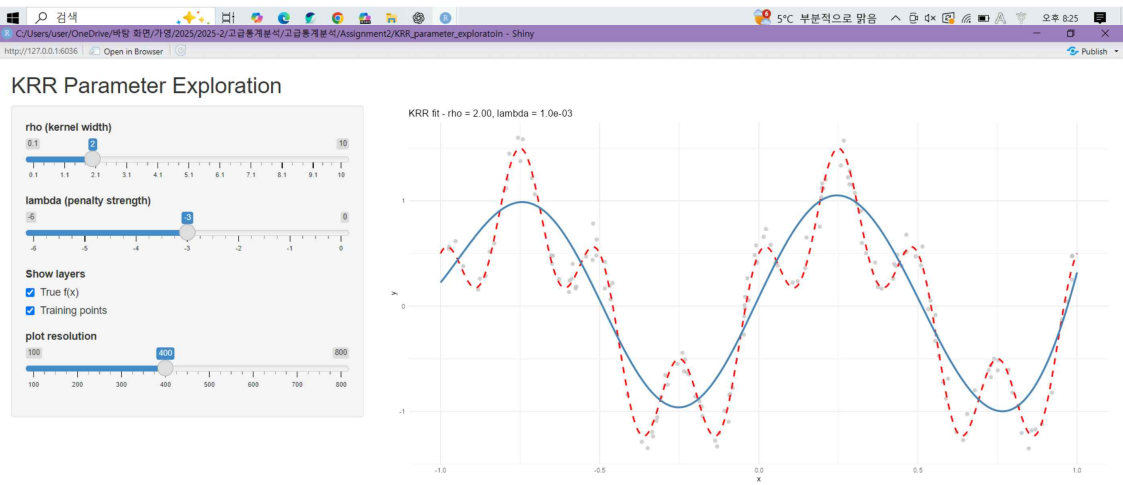
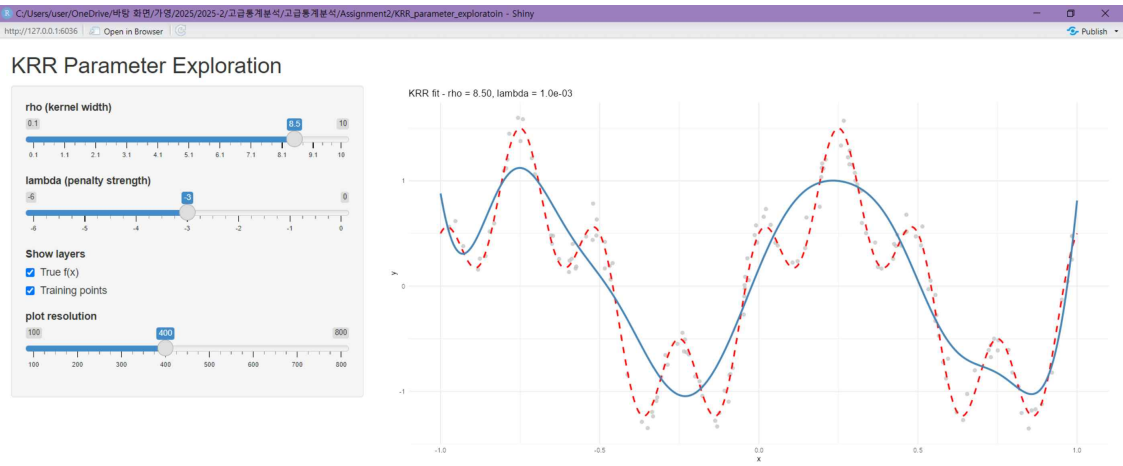
```
print(result)
```

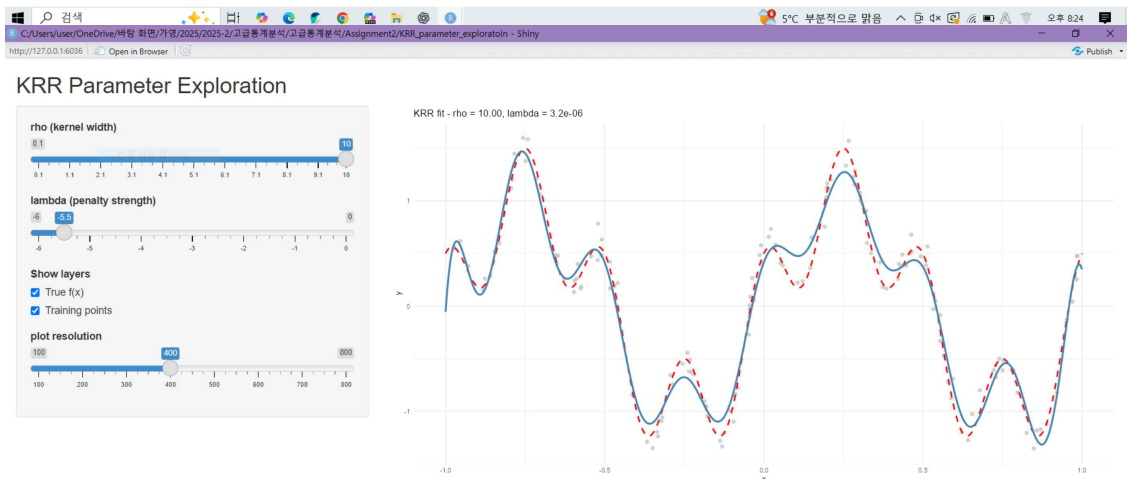
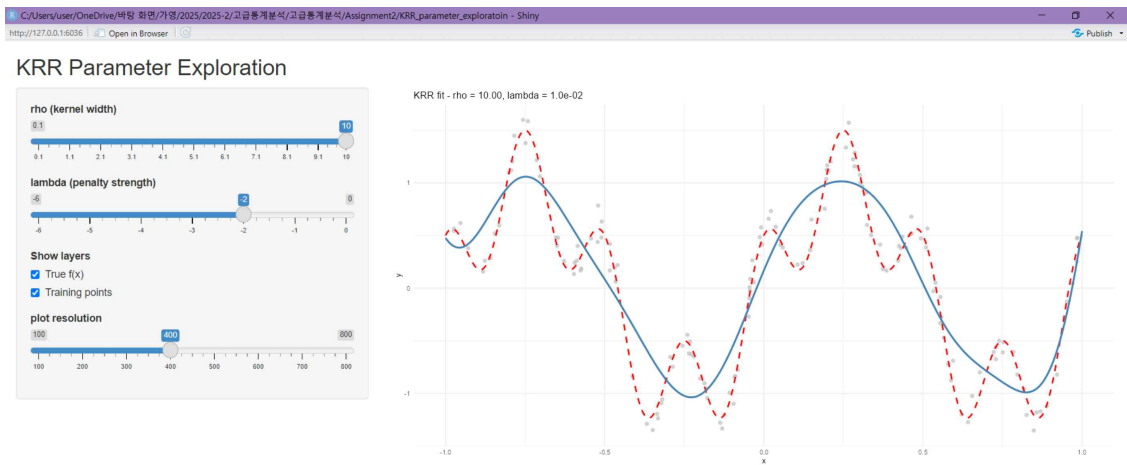
```
##   Method mean_time
```

```
## 1      R      0.1198
```

```
## 2    Rcpp      0.0630
```

Probelm 3.
(c).





→ rho 값은 그래프의 부드러운 정도를 정하고, lambda 값은 그래프가 데이터를 따라가는, 즉 적합하는 정도를 결정한다. (그렇기 때문에 과적합이 일어나지 않기 위해 패널티 항 앞에 lambda 파라미터를 사용한다.)